Week 5 – Creativity in Research and Ethics in Research (Lean Notes)

Lectures covered: L21–L23 (Creativity in Research, Parts 1–3, Dr C. Balaji) and L24 (Ethical Practices in Research, Dr Sreekumar Nellickappilly). ⭐ = tested in the Week 5 assignment.

1. Concept cards (in the order taught)

Part A: Creativity in research (L21–L23)

1. What a PhD is ⭐

- Joke graph of "how much you know": B.Tech = a little about everything; Masters = reasonable about many things, with doubts; **PhD = a lot about a very narrow area**; fully professional researcher = "knows completely about nothing" (a joke on how narrow the depth gets).
- Meaning: you go **deep** into one problem; through depth you gain **breadth** later (you can follow a talk from another department). **A PhD is deep, not broad.**

2. Learning curve

- Cooling of a hot body is an exponential decay (fast at first, then flat). **Learning is the opposite**: explosive at first, then flattens after 6–8 years, "time to go".

3. Attributes of a fully professional researcher ⭐

- **Peers want to hear you**: invited as keynote speaker at international meetings; **peer group truly international** (not just your city); you are among the top 10–20 in your area. Takes years ("Rome was not built in a day").
- **Nobel Prize is not for upstarts**: the work must survive 20–30 years ("time is a violent torrent"); radically different results from others would mean the issue is still open.
- **Mastery of techniques**: analytical, numerical and experimental (do not put all eggs in one basket; analytical-only problems are shrinking; software like Ansys changes; flow over a flat plate is no longer a hot topic).
- A **line of investigation** for the next 5–10 years (where are the gaps, where can you contribute?).
- **Communication**: papers, posters, talks. Read widely, even fiction and non-fiction, to learn styles. (iPhone is "creative" because people accepted it.)
- Guide students; be a **fountainhead of ideas for a lifetime**; know other fields; passion and hard work.
- In industry: write and defend proposals to your boss (state your USP); be among the top 10 in your area.

4. Attributes of a good researcher ⭐

- **Sound grasp of fundamentals**; **mastery of techniques** in the field; **tenacity and perseverance** (the experiment will not work, the program will not converge, the paper may be rejected); **scholarship** = deep knowledge of your problem plus **awareness of what others, including the top researchers, are working on** ("do not be a frog in a well"; today's problem will be settled in 3–5 years, so leave the comfort zone); **communication skills**.
- Be **well read in many areas**; **aware of recent advances**.

5. What creativity is

- Doing something **never done before**. It is not only gadgets: discovering penicillin, the law of gravitation, fluidised bed combustion, composing music (Mozart, A. R. Rahman), a new legal argument, a new reading of history. **Creativity appears in drama, science fiction and engineering, all of the above.** ⭐

6. Sternberg's triarchic theory of intelligence (1985, *Beyond IQ*, Yale) ⭐

Type	Meaning	Where you see it
Analytical	solve **structured problems that have one right answer**; test of fundamentals	classes, exams, grades, degree
Synthetic = creative (the lecturer says "synthetic"; the assignment says "creative")	**combine what you learned in a new way**, **new ideas, new theories, new problems**; many possible answers	project work, **research**, design problems
Practical	**adapt to the real environment, "street smart"**, sell your ideas to funding agencies, managers, editors, reviewers; never taught, learned over years	everyday life, career

- Education drills only analytical intelligence; the other two must be self-learned.
- Design example: rating a given heat exchanger (one answer) vs designing one for 100 kW (many answers).

7. Knowledge: why you cannot start research after Class 12

- Knowledge lets you **recognise what is new** (do not reinvent the wheel) and gives skills to **design experiments** (a clinical mercury thermometer cannot read 500 °C; use platinum resistance or infrared), **analyse results** (heat transfer coefficient, Nusselt number, efficiency) and do **scientific computing** (know the type of PDE and the methods). Have humility: your brilliant idea may already have been tried and failed.

8. Thinking style of creative people ⭐

- **Question conventional wisdom, assumptions and rules**, so they are **less likely to follow societal restrictions**, may clash with society, and are **more likely to find surprisingly new solutions**.
- Persistent, tenacious, stubborn; "arrogant" only in homeopathic dose (thin line between arrogance and self-confidence).

9. Hard work and the 10,000-hour rule

- About **10,000 hours of deliberate practice** to master a skill (tennis, music, research). Lecturer credits Herbert Simon (also Ericsson; popularised by Gladwell's *Outliers*, 2008).
- Tiny numbers: 3 h/day ≈ 10 years; 8 h/day full-time job ≈ **5 years** (hence 5 years as assistant professor, 5 years for a Master's); research student at **about 60 h/week ≈ 3 to 4 years** to gain the vision ("darshan"). Slip: he says "3 to 4 hours"; he means **years**.
- Talent alone is not enough: with above-average intelligence, hard work makes an expert. **Genius theory is debunked.**

10. Failure is normal

- Programs do not compile, then give garbage; take a break and the bug appears later. Even a professor with 450 papers has had rejections.
- Tiny number: if each of 3 reviewers accepts with probability 0.7, joint acceptance = $0.7 \times 0.7 \times 0.7 =$ **0.343**. So rejection is statistical; if the work is original, send it to another journal.
- Papers do not show the pain; society equates success with hard work, which is a **dangerous correlation** (low marks do not prove no study).

11. Motivation

- Motivation propels hard work. **Extrinsic** = reward, money, praise, promotion, fame. **Intrinsic** = your own goals, joy of solving an unknown riddle (the "Eureka" feeling); more important and more difficult; very high in highly creative people. Start with extrinsic, move to intrinsic.

- **Environment**: a good one helps but is not necessary (Ramanujan did great mathematics in adversity). Find alternatives instead of waiting for the ideal machine.

12. Role of stress ⭐

- Progress needs "optimal unhappiness" with the present state. **Eustress** = good stress, brings out the best; **distress** = harmful stress.
- **Yerkes–Dodson curve** (1908, experiments on rats; performance vs pressure): **hypo-stress (too little)** gives low performance, **eustress (moderate)** gives the **best**, **hyper-stress (too much) and distress** bring performance down again. Each person must find their own eustress point.

13. How creativity happens

- Often a **sudden intuitive flash of insight** ("darshan") outside the lab; **write it down at once**, then work hard to prove it. Example: **Max Planck 1901** found that putting "−1" in the black-body formula matched experiments; he then derived **E = nhv** (energy in multiples of hv; h = 6.627×10^{-34} J·s) and got the Nobel Prize in 1918 (about 17 years later). It started quantum mechanics.
- Many "brilliant ideas" fail: set a time limit (a month), then switch to a **backup** line.
- **Research progress follows a log-sigmoidal curve**: slow for the first 1–3 years (learning, set-up not working), then a rapid jump, then diminishing returns ("do not milk it dry with the sixth paper").

14. Rephrase the question: reductio ad absurdum

- Prove √2 is irrational by assuming it is rational (p/q) and reaching an absurdity. In research, look for alternative views and **new performance metrics**.
- **Do not state the obvious** (Nusselt number rises with Reynolds number). "Dog bites man" is not news, "man bites dog" is: aim for surprising, correct results (a smaller tube gives more heat transfer per unit pressure drop).

15. Choosing the right problem: Medawar zone

- Too easy (inverse of a 3 × 3 matrix) gives no payoff; too hard (prove God does not exist by thermodynamics) gives no payoff. **Peter Medawar** (Nobel 1960): payoff vs difficulty peaks in the middle, the **Medawar zone**. Right problem = 3–5 years, a good answer, a couple of papers, a PhD.
- Joy of research is intrinsic: "from darkness to light".

16. Examples of creativity in L23

- **Velcro**: George de Mestral turned burrs sticking to his trousers (a nuisance) into a product.
- **Post-it** (3M, 1974): Art Fry used Spencer Silver's "failed" weak adhesive as a removable bookmark by ignoring the belief that adhesives must hold permanently.
- **Bullet train**: nose shaped like a **kingfisher's beak** cut tunnel noise (bio-inspired).

17. Flow theory (Csikszentmihalyi, Chicago)

- Plot **challenge vs skill**: high skill + low challenge = **boredom**; low skill + high challenge = **anxiety**; matched = the **flow channel** (best performance, enjoyment, timelessness). Tennis example: wall practice is boring after two months; playing the champion on day one scares. Keep challenge growing with skill. Sign of flow: students do not look at their watches.

18. Managing creativity and age

- Accidental discoveries need extra time: advisers should give a month or two to chase an unexpected result. Intelligent people can filter which ideas are worth pursuing.
- Two kinds of scientists: **competent** (right place, right time) and **true genius** (generates many new ideas, e.g. Einstein). **Young scientists are fearless**, with no reputation to protect; after 10 years in a field, familiarity reduces curiosity.
- **Change fields about every 10 years** (Chandrasekhar, Nobel 1983: white dwarfs 1929–39, stellar dynamics, ellipsoidal figures, black holes, gravitational waves).

19. Why designs and research are not creative; traits of the creative engineer

- Resisting new ideas; lack of integrity; poor knowledge of fundamentals; not considering several ideas together (the machine with no ON/OFF switch); lack of **discernment and selectivity**, so **keep it simple** (Edward de Bono wrote on simplicity: the hardest goal is to simplify).
- Positive traits: **self-confidence** (remember the 10,000 hours), **high motivation**, **constructive nonconformity** (do genuinely different things, not look different), and the **ability to think in images** (drawing is the language of engineers).

20. Checklist to generate ideas

- Change or rearrange something? Increase or decrease something? Make moving parts stationary? Look at **asymptotes** ($Re \to 0$, $Re \to \infty$) to benchmark; do **paper-and-pencil scale analysis** first; deny the problem exists and take 10 days off; deny a natural law; try fantasy; **brainstorm**; break into sub-problems.
- **Analogy**: electrical resistance network for conduction; **nature** (biomimicry); ideas from other fields (computer science, AI) applied to yours.
- **Roadblock**: fixity or mental block.

21. Nature-inspired algorithms (worked example)

- **Ant colony optimisation**: ants leave **pheromone** on paths; later ants prefer high concentration, so the shortest path emerges (travelling salesman problem).
- **Genetic algorithm**: encode variables as 0s and 1s ("chromosomes"), pick fit parents, mix (crossover), randomly flip bits (**mutation**), check fitness; evolution as optimisation.
- Worked case: split 15 W among 15 heat-generating chips to **minimise the maximum temperature (minimax)**. Baseline = 1 W each. Use CFD to train a **neural network surrogate** (fast), run the genetic algorithm on it, then confirm in a wind tunnel (error under 10%, about 7 °C). Optimal powers are not equal (0.5, 1.091, 2 W…).

22. Summary of the creative process

- Creativity lies in the **generation of ideas**; building the setup comes after, and the value of results depends on the **quality of ideas**.
- Process: **soak the mind with information**, try combinations, solution appears; a dialogue between **conscious and subconscious** (even in dreams); build a **large library of ideas**, read biographies, talk to others, develop 3D visualisation.
- **Improve creativity** with games and puzzles, hobbies, poetry, a daily journal, short stories, writing about your experiences.
- Research is a serious, lifelong activity: like a compass, the mind should return to research. Success is judged by others and time (Prandtl's 1905 boundary-layer theory is still taught). **Take-home: problems cannot be solved by the same level of thinking that created them.**

Part B: Ethics in research (L24)

23. Research process and why ethics enters every step

- Problem → literature review → objectives and hypothesis → methodology (data collection, analysis) → results → publishing. Ethics matters from choosing the problem to publishing (authorship, plagiarism).

24. What ethics is

- From Greek *ethos* = **character**; the moral integrity of the researcher. Research depends on **public trust**, so researchers must be responsible. Professional bodies (engineering, medicine) have their own codes.

25. Ethical dimensions

- Problem selection; relations among researchers (collaboration, shared labs, publishing); **treatment of human and animal participants**; **truthfulness and transparency**; **risk management (be risk aware, not risk**

averse); **willingness to share benefits** (authorship, patents, payments to subjects); **act with responsibility**, which is the key word of the lecture.

26. Case: Hwang Woo-suk (Korea)

- Claimed in *Science* (2004) to have cloned a human embryo and extracted stem cells. A panel found the data **fabricated** (11 patient-specific stem cell lines false; DNA matches faked), and **female researchers in his lab were pressured to donate eggs**. He was dismissed in March 2006 and charged with fraud and embezzlement.
- Violations: **coercion, data fabrication, dishonesty, lost trust, lack of integrity, cheating**.

27. Other ethical violations ⭐

- **Plagiarism** (presenting others' work as yours without acknowledgement); **unfounded charges** against another researcher; **breach of confidentiality** (participants trusted you); **discrimination** (gender, caste, race, nationality, language); **negligence**; untruthful data management; harm to human or animal participants; insensitivity to cultural, legal and political context.
- **Why it happens:** research is frustrating, so people seek an easy way out; self-interest; **ignorance**; **relativism** (so collaborators must agree on practices beforehand); **pressure to publish ("publish or perish")** and win grants; ambition, profit, fame; **poor supervision of students**; insensitivity to harm.

28. Responsibility

- **Negative:** prevent harm of three types: **intentional, negligent, reckless**. **Positive:** work benefits the public, since your career rests on public support.

29. History of research ethics

Milestone	Point
After World War II, **Nuremberg trials**, **Nuremberg Code 1949**	Nazi doctors' experiments on prisoners (Jews, gypsies) and euthanasia without consent; code stresses **voluntary consent**. **Ends do not justify means.**
Tuskegee Syphilis Study, 1932–1972, US Public Health Service, Alabama	only **black** men studied; **untreated** even after medicine existed; told "bad blood"; **no consent**; racial discrimination; stopped after media exposure
Declaration of Helsinki 1964	biomedical research on humans must weigh **risk against benefit**, respect privacy, minimise cost to subjects
Belmont Report 1979	respect for persons and **autonomy**
Animal Welfare Act (US)	care of research animals
Ethics committees, **Institutional Review Boards**	permission of the organisation's board is mandatory before starting

30. Human participants

- **Informed consent** (all information given; right to **withdraw or refuse** at any time); **confidentiality** (no names or addresses in publications without consent); **security** (data and samples secure, anonymised); **safety** (no unnecessary or disproportionate risk).

31. Animals

- Used as disease models, vaccine and drug development, **toxicity tests**. Issues: animals cannot consent; "we eat them anyway" is a weak argument; which species (mice, rats, monkeys; chimpanzees too close to humans?). Use only when necessary.

32. Obligations of researchers, and authorship ⭐

- **Honesty, integrity, minimal risk, cultural sensitivity, transparency, readiness to share benefits, social benefit above individual benefit, risk awareness, respect for intellectual property, responsible mentoring, no discrimination.**

- **Authorship:** credit by **merit**. **Substantial contribution to the ideas** → co-author. Only help here and there → **acknowledge**. Everyone who is an author must know about and agree to the publication. Publishing is for social benefit, not name and fame.

2. Top traps (most likely first)

1. **Sternberg:** practical = environment and street smart; creative = new theories and new problems (both options with "new" are creative); analytical = structured problems, exams. The lecturer says "synthetic", the assignment says "creative".
2. **PhD makes you a specialist in one narrow area** (not broad, not fast, not a sketcher).
3. **Eustress** (moderate stress) gives peak performance; hypo-, hyper- and distress give less. Yerkes–Dodson on rats, 1908.
4. **Creative people** are less bound by social rules and more likely to give surprising solutions; they do not stay within previous work.
5. **Success measure:** peer recognition at the international level and peer-reviewed publication (campus magazine or department meeting do not count).
6. **Collaborative paper:** every co-author must know and agree; being first to type it, junior or senior status is irrelevant.
7. **Good researcher / professional researcher:** communication, technique mastery, line of investigation, awareness of recent advances; command of the area; where to contribute; **never ignores literature**.
8. **Widely read** researchers connect ideas; creativity is not only in arts.
9. Lecturer slips: 60 h/week gives "3 to 4 **years**" (not hours); Simon's 10,000-hour rule is popularised by Ericsson and Gladwell.
10. **Result of three reviewers:** joint acceptance is the product ($0.7^3 = 0.343$), not the average.
11. **Ethics:** Nuremberg (consent), Tuskegee (untreated, race), Helsinki (risk–benefit), Belmont (autonomy). Responsibility has two sides (avoid harm, benefit the public). Be risk aware, not risk averse.
12. **Authorship by merit:** substantial contribution → co-author; minor help → acknowledgement.

3. Assignment questions with solutions

Q1. According to Sternberg's theories of intelligence, practical intelligence relates to (select all that apply) *(Q1)* - A. How one relates to the environment around them B. How street smart they are C. How well they come up with new theories D. How well they solve new problems

Solution (card 6): Practical = adapting to the real environment and street smartness. New theories and new problems are creative intelligence. **Answer: A and B.**

Q2. A few research scholars collaborated on some area, generated data and analysed the results. Some of them published the result independently since they were the first to type it up. Is it ethical? *(Q2)* - A. Yes, because they have collaborated and know each other B. No, all authors should know and agree with the publication C. Yes, if the collaborators are junior to you D. No, only if the authors are senior to you

Solution (card 32): All authors must know about and agree to the publication, whatever their seniority. **Answer: B.**

Q3. Creativity is seen in *(Q3)* - A. Drama B. Science fiction C. Engineering D. All of the above

Solution (card 5): Creativity appears in every field, not just arts. **Answer: D.**

Q4. A good researcher is often well read in many areas. *(Q4)* - A. True B. False

Solution (cards 3, 4): Reading widely lets you bring ideas from other fields (scholarship). **Answer: A (True).**

Q5. We can work to the best of our potential when we are *(Q5)* - A. Hypo stressed B. Eustressed C. Hyper stressed D. Distressed

Solution (card 12): Peak of the Yerkes–Dodson curve is moderate (good) stress. **Answer: B.**

Q6. The measure of a successful researcher is that the work should be (select all that apply) *(Q6)* - A. Recognised by peers at an international level B. Typically published in peer-reviewed journals C. Mentioned in the campus magazine of the college D. Must be discussed in the department meeting

Solution (cards 3, 4): Success means an international peer group listens to you, and work is accepted through peer review. **Answer: A and B.**

Q7. Creative people are (select all that apply) *(Q7)* - A. More likely to follow societal restrictions B. Less likely to follow societal restrictions C. More likely to think within the limits of previous work D. More likely to come up with a surprisingly new solution to a problem

Solution (card 8): They question rules and assumptions. **Answer: B and D.**

Q8. A PhD usually makes you *(Q8)* - A. A specialist in one narrow area B. A person who conducts experiments fast C. A person with wide knowledge of many topics D. A person who makes excellent sketches

Solution (card 1): Depth in one narrow problem. **Answer: A.**

Q9. To become a good researcher, one must usually (select all that apply) *(Q9)* - A. Have good communication skills B. Have mastery over specific techniques C. Have a line of investigation D. Be aware of recent advances in the field

Solution (cards 3, 4): All four were listed. **Answer: A, B, C and D.**

Q10. The following can be attributed to a fully professional researcher (select all that apply) *(Q10)* - A. They have fundamental concepts that international peers might be interested in B. They have command over their area C. They are aware of the situations where they can make a significant contribution D. They work without focusing on literature in the area

Solution (card 3): A, B and C match the professional attributes; ignoring literature is the opposite of scholarship. **Answer: A, B and C.**